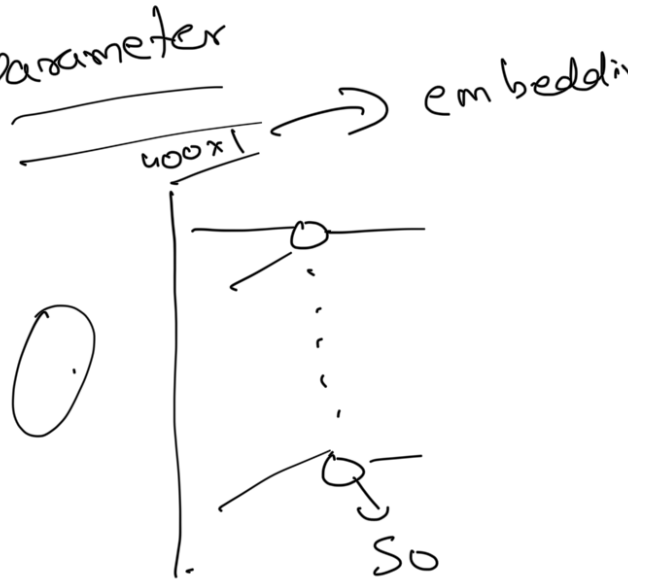
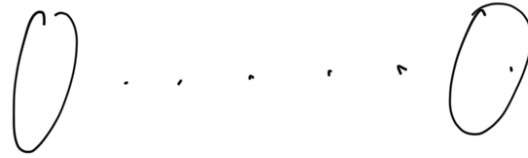


Vector representation of input data ("Imo")
 Vector dimension will be a hyperparameter

Lenet



Lenet for Classification

1) A final layer with Softmax non-linearity

2) Number of neurons in final layer is equal to number of classes

Lenet for Embedding generation

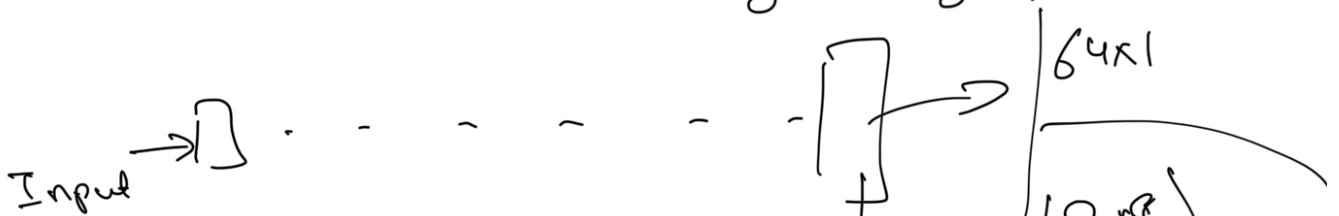
1) A final layer with ReLU and tanh

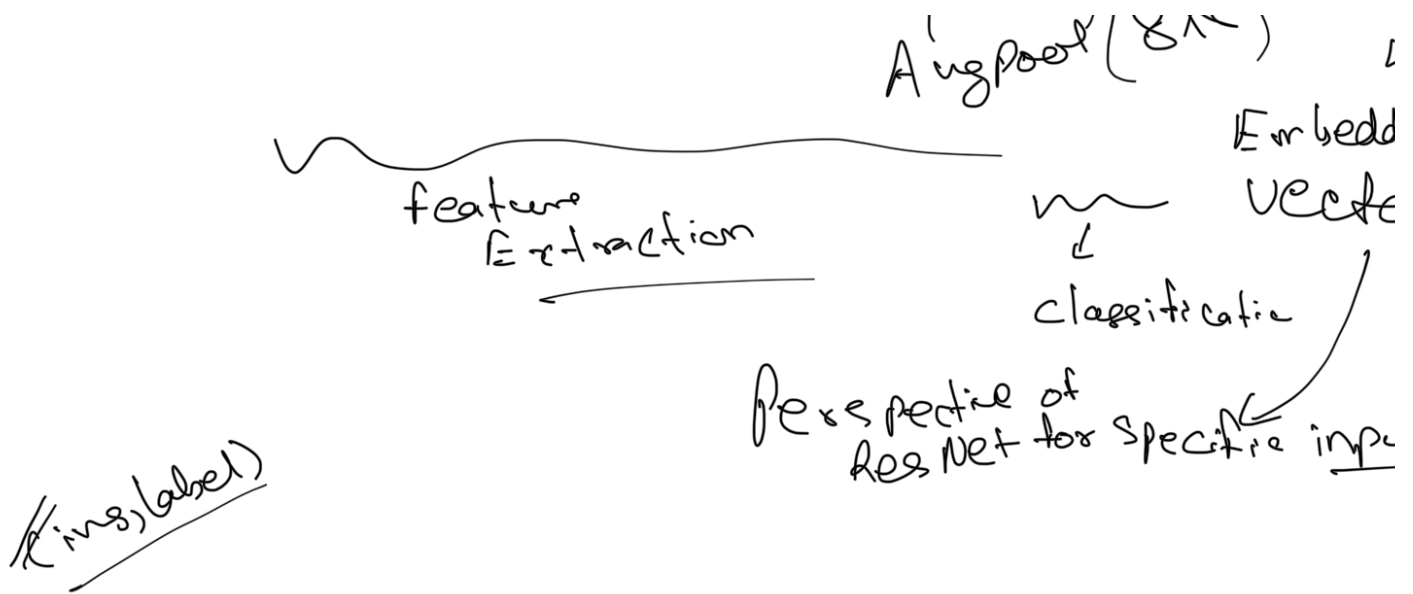
Number of neurons is equal to the required embedding dimension (hyperparameter)

flattening

Embedding is like a perspective of the neural network

Embedding using Resnet





LAB on Siamese Network

Step 1: Prepare Dataset in $(img_1, img_2, label)$ format

class

label $\begin{cases} 0 & - \text{similar} \\ 1 & - \text{dissimilar} \end{cases}$

Step 2: Create deep learning network

class

Step 3: Create siamese network with two branches; where each branch has the network designed in step 2.

class

Step 4: Contrastive loss to evaluate distance between two images

class

$$L(y, d) = \frac{(1-y)d^2}{2} + y(\max(0, m-d))^2$$

label $d(A, B)$

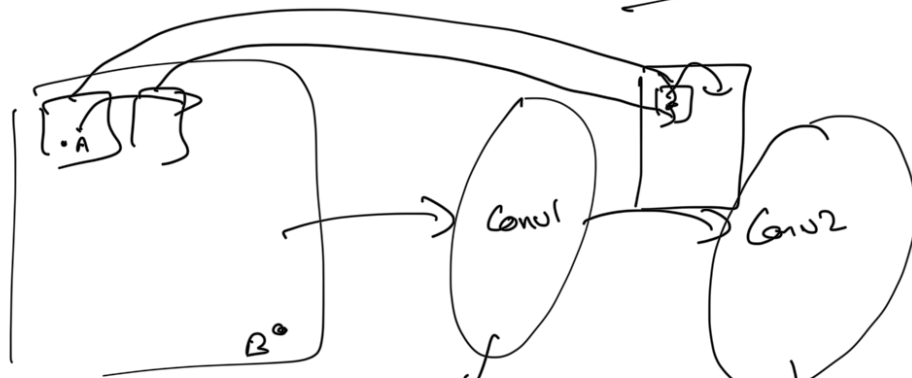
Problems in CNNs

80-85%

X

Inductive Bias

Initial layers of the network are focusing only on a very small local neighbourhood

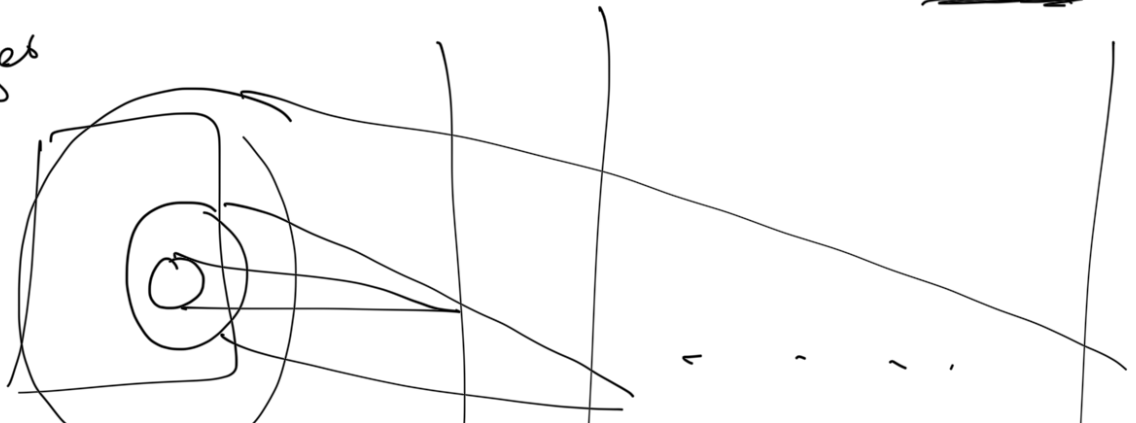


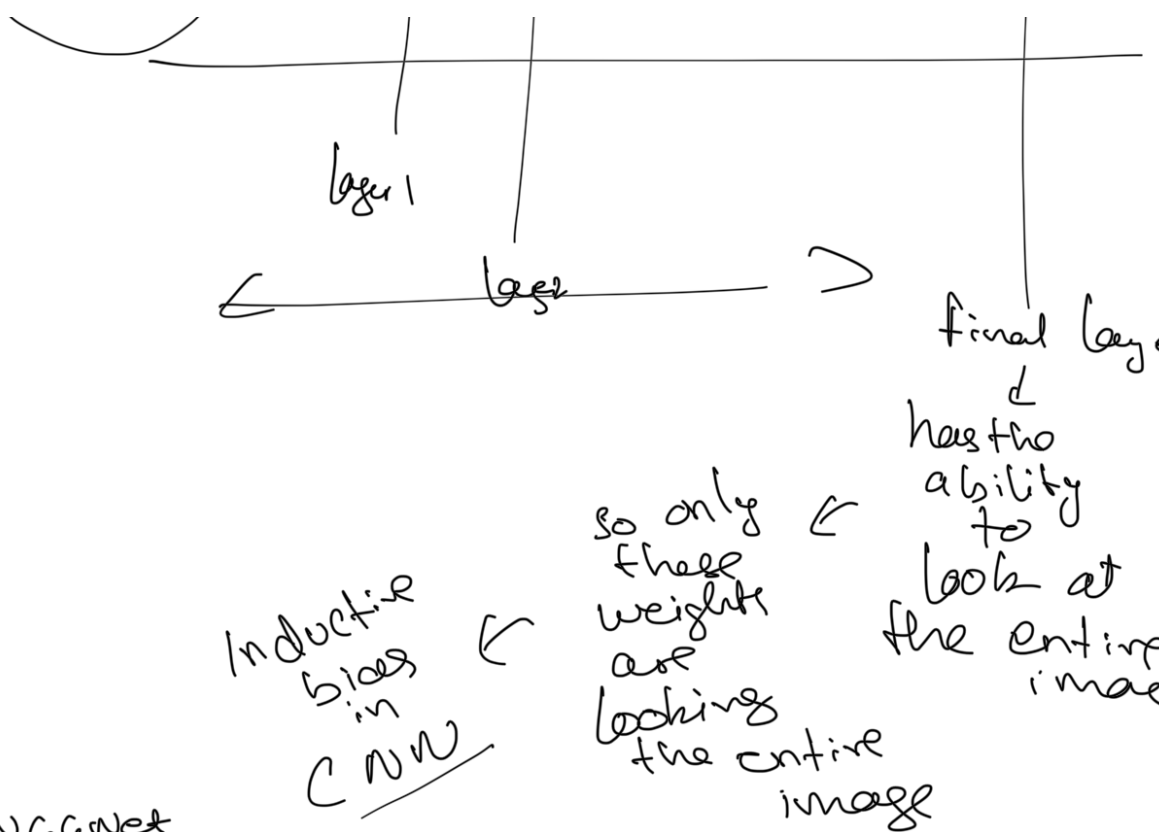
Application is localized to kernel width

Application of layer is localized but is larger layer 1



Increasing vision by layer layer





2015 VGGNet

will try to resolve this problem of inductive bias in CNNs

Transformer Models

2017.
Transformer for Natural Language Processing

NLP Application: Next 'word' Prediction
Using Transformer models
Classification
↓ then

because and

